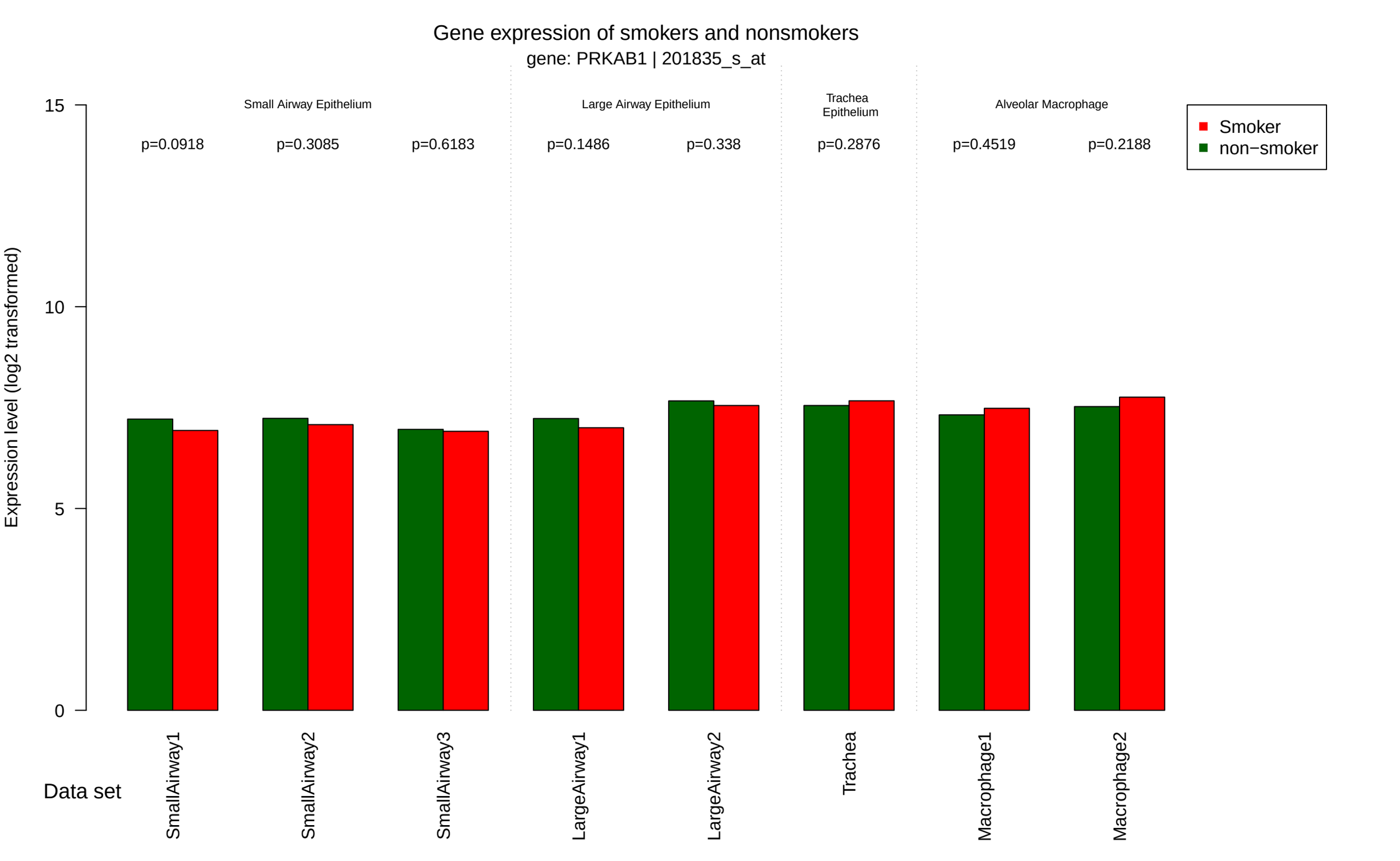
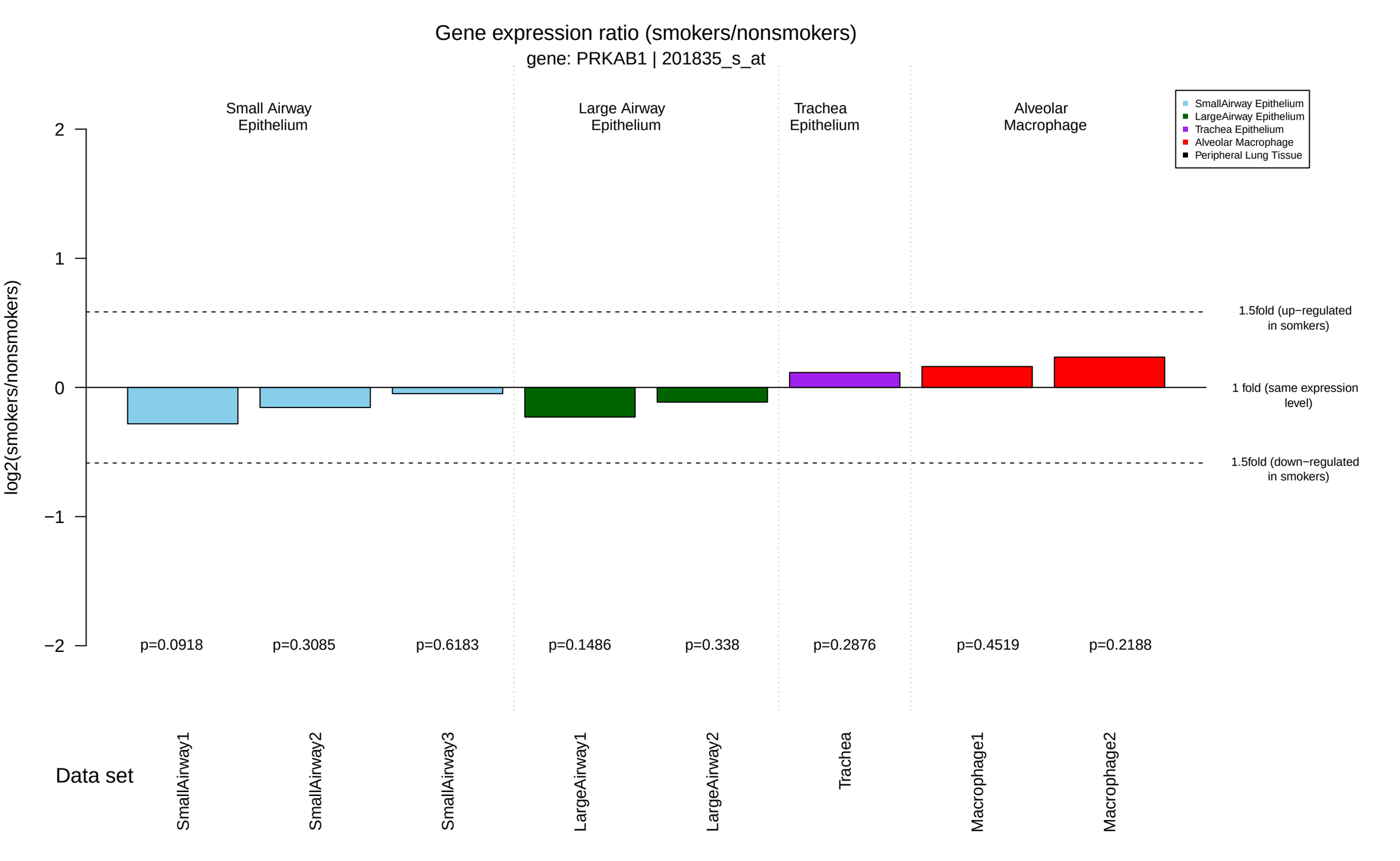
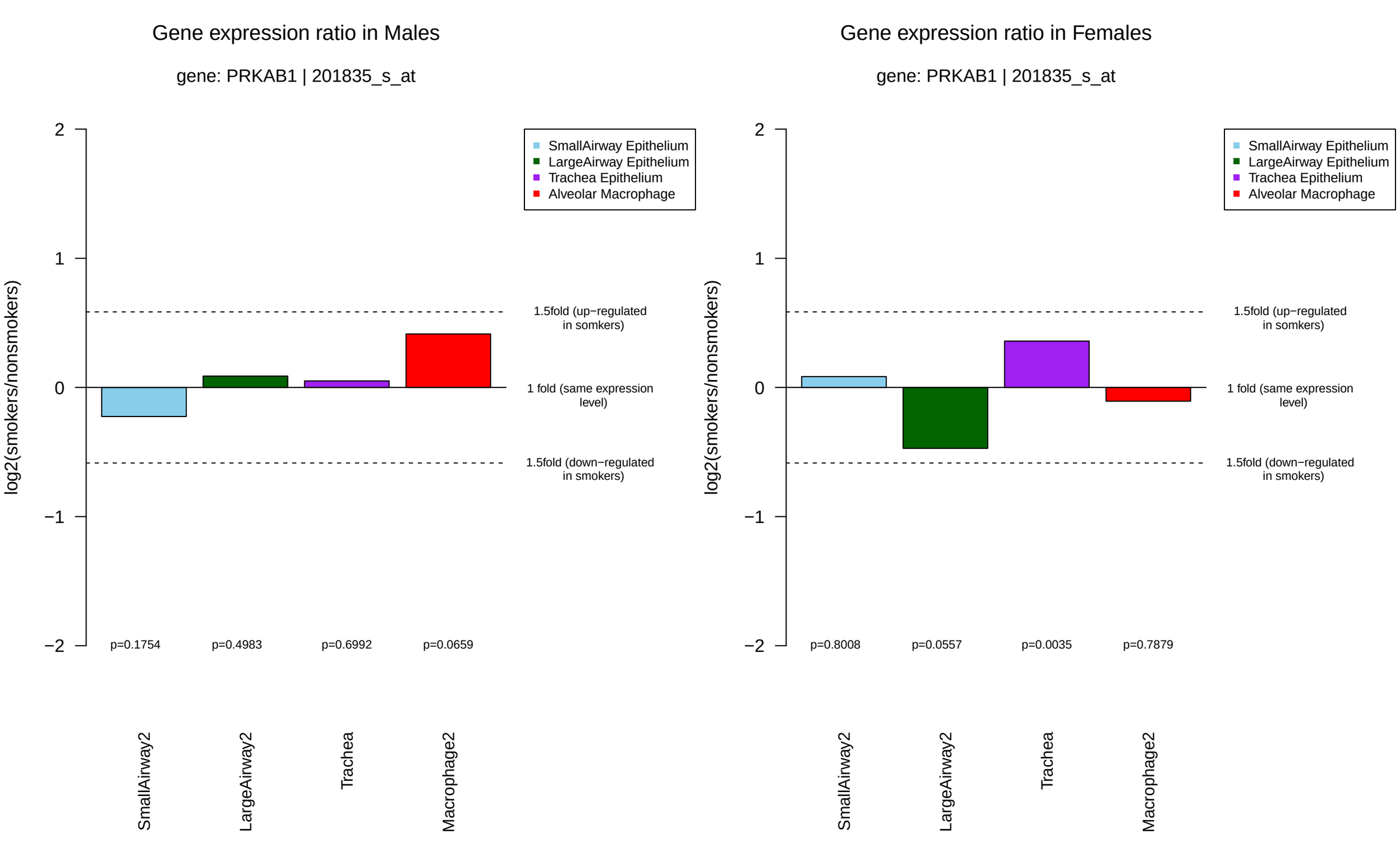
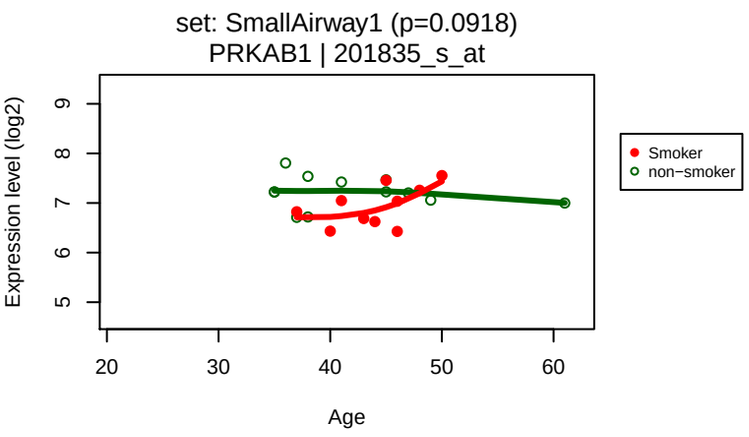
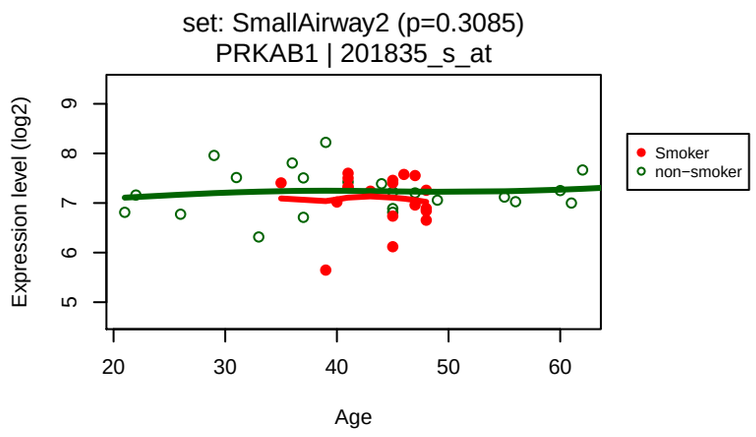
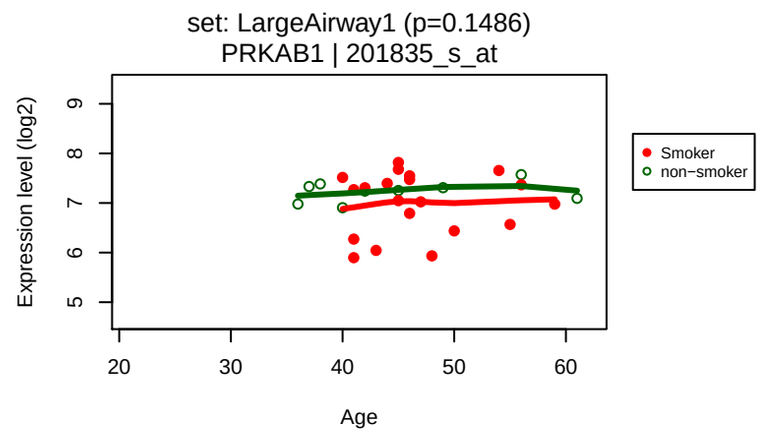

Gene expression of smokers and nonsmokers
gene: PRKAB1 | 201835_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0918
p=0.3085
p=0.6183
p=0.1486
p=0.338
p=0.2876
p=0.4519
p=0.2188
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

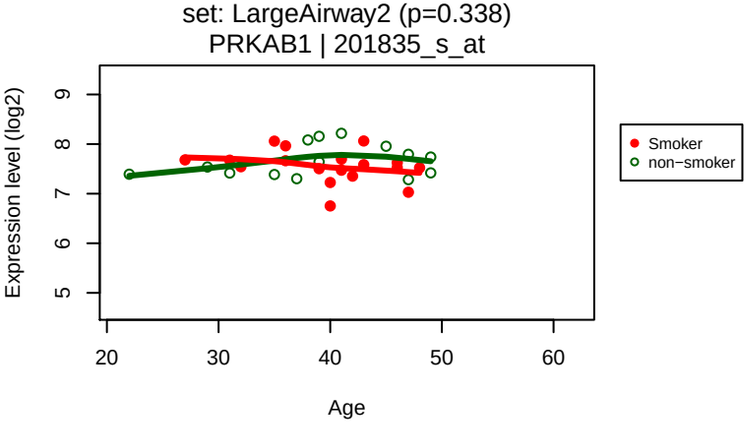

Gene expression ratio (smokers/nonsmokers)
gene: PRKAB1 | 201835_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0918
p=0.3085
p=0.6183
p=0.1486
p=0.338
p=0.2876
p=0.4519
p=0.2188
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

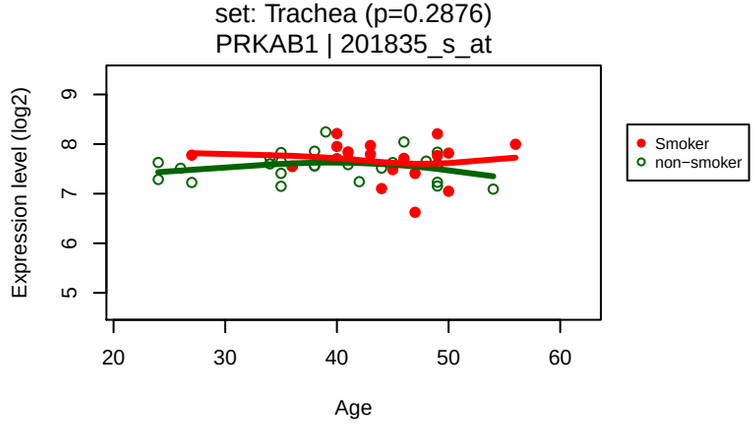

Gene expression ratio in Males
gene: PRKAB1 | 201835_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1754
p=0.4983
p=0.6992
p=0.0659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAB1 | 201835_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8008
p=0.0557
p=0.0035
p=0.7879
SmallAirway2
LargeAirway2
Trachea
Macrophage2

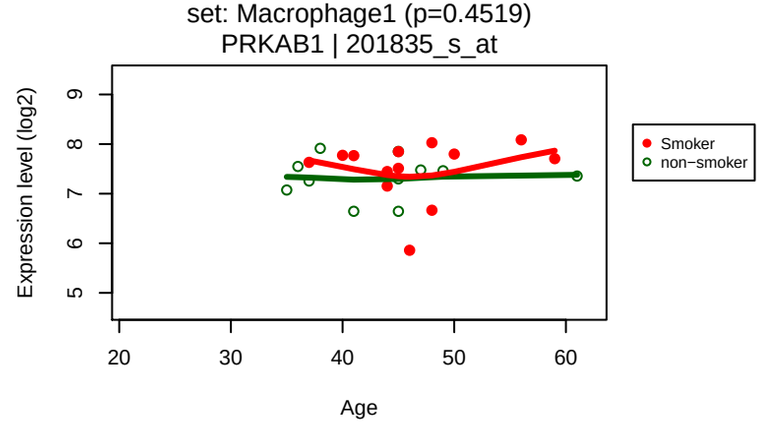

set: SmallAirway1 (p=0.0918)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker

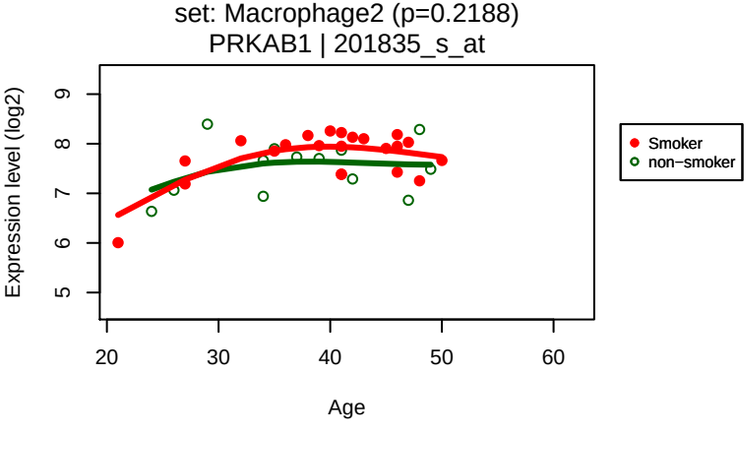

set: SmallAirway2 (p=0.3085)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker

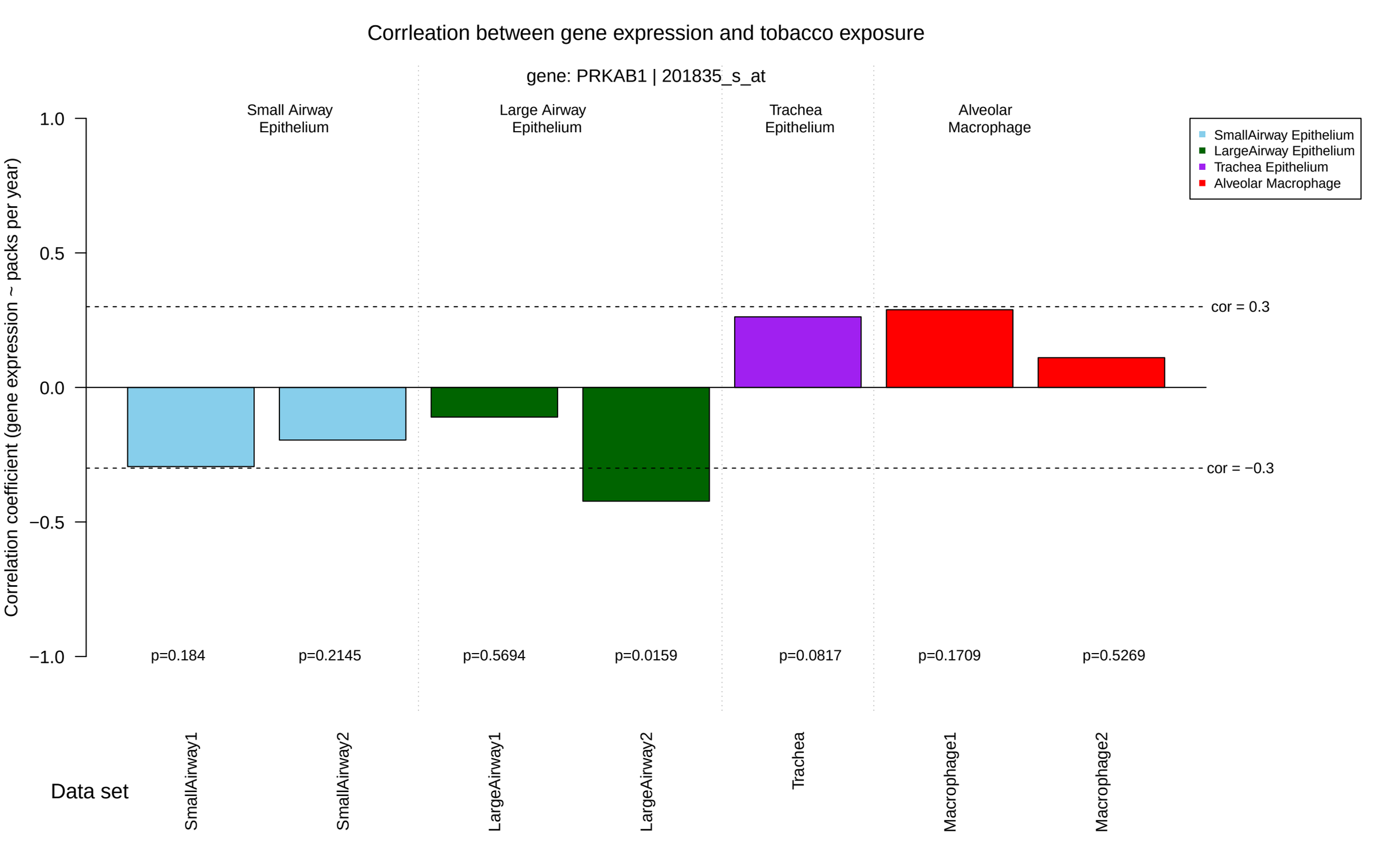

set: LargeAirway1 (p=0.1486)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.338)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2876)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4519)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2188)
PRKAB1 | 201835_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAB1 | 201835_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.184
p=0.2145
p=0.5694
p=0.0159
p=0.0817
p=0.1709
p=0.5269
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2